Title: Tree species diversity and structure mediate land surface phenology in seasonally dry tropical woodlands

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Author contribution statement

JLG conceived the study, conducted the analysis, and wrote the first draft of the manuscript. AS coordinated plot data collection in Zambia. All authors contributed to manuscript revisions.

Data accessibility statement

The data used in this study are from the Zambian Integrated Land Use Assessment (ILUA-II). Data were cleaned and archived by the SEOSAW project (Socio-Ecological Observatory for Studying African Woodlands). An anonymised version of the plot data are available at the following DOI: **URL for Edinburgh Datashare**. Code developed for data analysis can be found here: `https://github.com/johngodlee/zambia_phenology`.

Abstract

1. Seasonal foliage production (leaf phenology) is a key determinant of ecosystem function. Land surface phenology observed via space-borne remote sensing, can be predicted to some extent from climate, but we lack understanding of how vegetation structure and composition mediates patterns of land surface phenology. This hampers our ability to predict phenological changes and therefore ecosystem function, in light of rapid ongoing shifts in biodiversity and ecosystem structure.

2. We combined a dense network of 619 vegetation monitoring sites across deciduous woodlands in Zambia with land surface phenology metrics to investigate the role of tree species diversity, composition, and vegetation structure on patterns of land surface phenology, including the phenomenon of pre-rain green-up.

3. Tree species diversity was associated with earlier pre-rain green-up, a longer growing season, and greater cumulative foliage production, across all woodland types identified in the study. Proportional abundance of Detarioideae species (subfamily of Fabaceae) drive a longer period of maximum foliage production by facilitating earlier pre-rain green-up. Woodland compositional types differed maginally in their land surface phenology. Larger trees can access deep groundwater reserves, facilitating earlier pre-rain green-up. Senescence metrics showed consistent variation among sites, but were not well explained by any measured variable.

4. This study underlines the importance of vegetation composition, diversity and structure as determinants of ecosystem function, and offers new insights into the factors determining patterns of land surface phenology across the dry tropics, which are essential for realistic modelling of land-atmosphere mass and energy exchange.

5. **Synthesis:** Tree species composition and diversity influence seasonal foliage production in deciduous tropical woodlands. More diverse woodlands are able to green-up earlier with respect to seasonal rains. Detarioideae species (subfamily of Fabaceae), are associated with longer growing seasons, effectively extending the period of maximum foliage production by facilitating pre-rain green-up. Understanding variation in land surface phenology with respect to vegetation composition is of great importance for earth-system modellers aiming to parameterise the next generation of predictive models of land-atmosphere mass and energy exchange.

1 Introduction

Foliage forms the primary interface between the vegetated land surface, the atmosphere and incoming solar radiation (Gu et al., 2003; Penuelas et al., 2009). Seasonal cycles of foliage production (leaf phenology) play an important role in regulating global carbon, water and nitrogen cycles (Richardson et al., 2013). Terrestrial biosphere models routinely incorporate vegetation indices derived from remote sensing products as a proxy for foliage production (land surface phenology), to constrain estimates of Gross Primary Productivity (GPP) (Bloom et al. 2016; Helman 2018). Previous studies have described environmental drivers of land surface phenology (Adole et al., 2019; Guan et al., 2014), but understanding of how the composition and structure of the vegetation itself affects patterns of land surface phenology is lacking (Whitley et al., 2017).

At continental scales, land surface phenology can be predicted adequately using only climatic factors, namely precipitation, diurnal temperature oscillation, and photoperiod (Adole et al., 2018b; Adole et al., 2019; Guan et al., 2014), but significant local variation exists within biomes in the timing of foliage production that cannot be attributed solely to abiotic environment (Stöckli et al., 2011). It has been repeatedly suggested that the diversity, composition, and structure of plant communities plays a role in determining ecosystem response to abiotic cues driving patterns of land surface phenology (Adole et al., 2018a; Jeganathan et al., 2014; Fuller, 1999), owing to differences in phenological strategy among species. Indeed, ground observations find wide variation in phenological patterns of foliage production among plant communities within a given biome (Seyednasrollah et al., 2019), but this is rarely represented in predictive models of biosphere-atmosphere exchanges (Scheiter et al., 2013; Pavlick et al., 2013).

Across the dry tropics, seasonal oscillations in water availability drive seasonal cycles of foliage production (Chidumayo, 2001; Dahlin et al., 2016). The phenomenon of pre-rain green-up seen in some tree species within the dry tropics serves as a striking example of adaptation to seasonal variation in water availability (Ryan et al., 2017). Conservative species, slower growing with robust leaves and denser wood, may initiate foliage production (green-up) before the rainy season has commenced. They may also retain their leaves for longer after the end of the rainy season, though the mechanisms of this behaviour are less clear (Giraldo & Holbrook, 2011; Kushwaha et al., 2011). Acquisitive species however, may only begin to produce foliage during the rainy season, creating a dense leaf-flush during the mid-season peak of growth and dropping their leaves earlier as soil water content drops towards the end of the rainy season (Lasky et al., 2016). While conservative species gain a competitive advantage from having fully emerged leaves once the rainy season starts, they must also invest heavily in deep root architecture to access groundwater reserves in order to produce foliage during the dry season, and risk hydraulic failure if seasonal rains occur much later than

normal (Vinya et al., 2018). It has been suggested that variation in phenological strategy among tree species is one mechanism by which increased species diversity increases resilience to drought and maximises productivity in water-limited woodland ecosystems (Stan & Sanchez-Azofeifa, 2019; Morellato et al., 2016).

Variation in seasonal patterns of foliage production affects broader ecosystem processes. Woodlands with a longer tree growth period support a greater diversity and abundance of wildlife, particularly birds, but also browsing mammals and invertebrates (Cole et al., 2015; Araujo et al., 2017; Morellato et al., 2016; Ogutu et al., 2013). As climate change increases the frequency and severity of droughts in water-limited woodlands, woodlands with a diverse tree community may provide refugia for many animal species (Bale et al., 2002). The period of green-up is a key time for invertebrate reproduction (Prather et al., 2012) and herbivore browsing activity (Velasque & Del-Claro, 2016; Morellato et al., 2016). Pre-rain green-up provides a valuable source of moisture and nutrients before the rainy season, and can moderate the understorey microclimate, increasing humidity, reducing UV exposure, moderating diurnal oscillations in temperature, and reducing ecophysiological stress which otherwise can lead to mortality during the dry season. Thus, understanding what drives variation in seasonal patterns of foliage production in tropical deciduous woodlands can provide valuable information on their vulnerability to climate change, and therefore better information to predict their future extent and productivity.

In the miombo woodlands of southern Africa, the largest woodland type in the region (White, 1983), Detarioideae species (subfamily of Fabaceae) are one such group which exhibit pre-rain green-up. Detarioideae species are frequently dominant in miombo woodlands, producing tall woodland canopies with deep root systems (Zhou et al., 2020). Previous studies have shown local variation in land surface phenology, particularly the pre-rain green-up phenomenon, which is not explained by climate (Zhou et al., 2020).

In this study we investigate how tree species diversity, composition, and structure influence land surface phenology in seasonally dry tropical woodlands. We focus specifically on the lag between green-up/senescence and the onset/end of the rainy season, the magnitude of foliage production within the growing season, and the overall length of the growing season. We hypothesise that: (H_1) sites with greater species diversity will exhibit a longer growing season due to a higher diversity of phenological strategies; (H_2) in sites with greater species diversity the start of the growing season will occur earlier with respect to the onset of the rainy season due to an increased likelihood of containing species which can green-up early; (H_3) sites with larger trees will exhibit longer growing seasons, as large trees can better access deep groundwater reserves outside of the rainy season; (H_4) sites dominated by Detarioideae species will experience earlier pre-rain green-up.

2 Materials and Methods

2.1 Plot data

We used data on tree species diversity and composition from 619 sites surveyed as part of the Zambian Integrated Land Use Assessment Phase II (ILUA-II), conducted in 2014 (Mukosha & Siampale, 2009; Pelletier et al., 2018). Each site consisted of four 20×50 m (0.1 ha) plots positioned in a 500×500 m square around a central point (Figure 1), resulting in a total plot area per site of 4000 m^2. The original census contained 993 sites, which was filtered in order to define study bounds and to ensure data quality. Only sites with ≥50 stems ha^{-1} ≥10 cm DBH (Diameter at Breast Height) were included in the analysis, to ensure all sites represented woodlands with a relatively continuous canopy rather than 'grassy savanna', which is considered a separate biome with different species composition and ecosystem processes governing phenology (Parr et al., 2014).

Plots in the ILUA-II dataset sampled stems ≥10 cm DBH across the area of the whole plot, while stems <10 cm DBH were measured in 10×20 m (0.02 ha) nested subplots. We chose to use only stem measurements ≥10 cm DBH in this study, discarding measurements from the nested subplots. In woodland ecosystems with a relatively continuous canopy like those studied here, only the larger trees form the canopy layer, with smaller trees existing within the understorey layer (Chidumayo, 2001). The signal from remotely-sensed measurements of land surface phenology is dominated by these larger trees which form the woodland canopy. We therefore reasoned that small stems would not be an important factor in determining measurements of land surface phenology. To further demonstrate the dominance of larger stems in our sites, we compared the total basal area contribution of stems ≥10 cm with that of stems <10 cm DBH, with the values of the small stems multiplied up to the area of the whole site. We found that across all sites, the minimum proportional basal area contributed by stems ≥10 was 72%, while the median was 98%.

Sites with non-native tree species, e.g. *Pinus* spp. and *Eucalyptus* spp. were excluded, as these species may exhibit patterns of foliage production which differ markedly from that of native tree species assemblages (Broadhead et al., 2003). Of the 49656 trees recorded, 97.9% were identified to species. Sites with >20% of trees not identified to species were discarded.

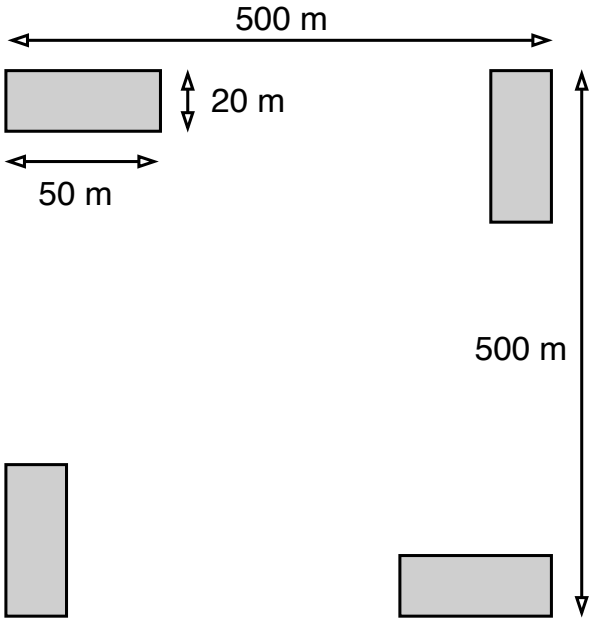

Figure 1: Schematic diagram of plot layout within a site. Each 20×50 m (0.1 ha) plot is shaded grey. Note that the plot dimensions are not to scale.

Within each plot, the species and basal area of all trees with at least one stem ≥10 cm DBH were recorded. Plot data were aggregated to the site level for analyses to avoid pseudo-replication. Each site spans an area 500×500 m (0.25 km^2), matching that of the land surface phenology data used. We excluded sites where plots differed appreciably in vegetation composition, to ensure that all sites used in statistical analysis were representative of the vegetation within the total 0.25 km^2 area of each site. Using the Bray-Curtis dissimilarity index on species basal area data (Faith et al., 1987), we calculated the mean pairwise compositional distance between plots within each site. Sites were discarded when the mean pairwise compositional distance among plots within the site was greater than the mean pairwise compositional distance of all plots across all sites. This resulted in a final dataset of 619 sites.

2.2 Vegetation composition and diversity

To classify variation in tree species composition we used agglomerative hierarchical clustering on species basal area data (Kreft & Jetz, 2010; Fayolle et al., 2014). We excluded species with fewer than five records across all sites from this analysis, as very rare species can hinder meaningful cluster delineation. We used Ward's algorithm to define clusters (Murtagh & Legendre, 2014), based on the Bray-Curtis distance between pairs of sites. We determined the optimal number of clusters by maximising the mean silhouette width among clusters (Rousseeuw, 1987). We described the vegetation types represented by each cluster using Dufrêne-Legendre indicator species analysis

(Dufrêne & Legendre, 1997), which ranks species as the product of their relative frequency and their relative average abundance in each vegetation type. Four vegetation types were identified during hierarchical clustering. The mean silhouette value of the clustering algorithm reached 0.64.

There was some spatial and climatic stratification in vegetation type distribution (Figure 2). Three miombo woodland types were identified, each with different sets of secondary species, but all dominated by Detarioideae species such as *Brachystegia* spp. and *Julbernardia* spp. (Table 1). Uapaca miombo is largely absent from the southwest of the country, occurring predominantly in higher rainfall regions to the north and east (Figure 3). Cryptosepalum miombo is found more in the southwest of the country, possibly representing the drier "Angolan miombo woodlands" described by White (1983). Importantly, we do not think these woodlands represent the very different Zambezian evergreen forests, which are dominated by Cryptosepalum. Julbernardia miombo is common in cooler areas and is more climatically restricted than the other miombo woodland types. Combretaceae woodlands match the description of small stature Zambesian woodlands, as described by Dinerstein et al. (2017) and Chidumayo (2001). These woodlands are not dominated by the same archetypal Detarioideae tree species as the other miombo woodland vegetation types. These Combretaceae woodlands occur over a wide climatic range, and contain some of the warmest sites in our dataset in the southeast of the country. The miombo woodland vegetation types are dominated by *Brachystegia* spp. and *Julbernardia* spp., with different secondary species. Median species richness and the range of species richness values per site is similar across vegetation types (Table 1).

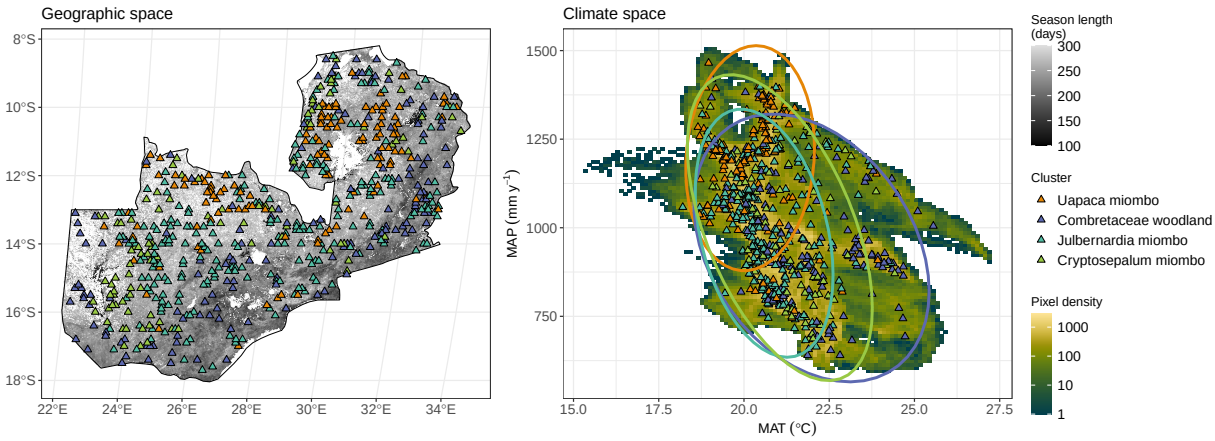

Figure 2: Distribution of study sites, within Zambia (left), and in climate space (right). Sites are shown as triangles, coloured according to vegetation type. Zambia is shaded according to mean growing season length during the period 2001 to 2021, estimated from the MODIS MCD12Q2 land surface phenology product, at 500 m spatial resolution (Friedl et al., 2022). Growing season length is masked by the MODIS MCD12Q1 land cover map from 2015 (Friedl & Sulla-Menashe, 2019), removing all pixels occurring in wetlands, croplands, water bodies, and urban areas. Climate space is represented by Mean Annual Temperature (MAT) and Mean Annual Precipitation (MAP), extracted from the WorldClim dataset at 30 arc second resolution (Fick & Hijmans, 2017). The shaded area in the right panel shows the climate space of Zambia, showing the density of pixels for given values of MAT and MAP. The ellipses in the right panel show the 95% confidence interval for modelled climate space of each vegetation type.

Cluster	N sites	Richness	Indicator species	Indicator value
Uapaca miombo	135	17(7)	*Brachystegia longifolia*	0.393
			Uapaca kirkiana	0.384
			Marquesia macroura	0.278
Combretaceae woodland	144	14(5)	*Combretum molle*	0.258
			Lannea discolor	0.229
			Combretum zeyheri	0.212
Julbernardia miombo	244	17(6)	*Julbernardia paniculata*	0.556
			Brachystegia boehmii	0.540
			Pseudolachnostylis maprouneifolia	0.227
Cryptosepalum miombo	96	14(6)	*Brachystegia spiciformis*	0.582
			Cryptosepalum exfoliatum	0.285
			Guibourtia coleosperma	0.282

Table 1: Dufrêne-Legendre indicator species analysis for the vegetation type clusters identified by the PAM algorithm, based on basal area weighted species abundances (Dufrêne & Legendre, 1997). The three species per cluster with the highest indicator values are shown alongside the median and interquartile range of site species richness and the number of sites within each cluster.

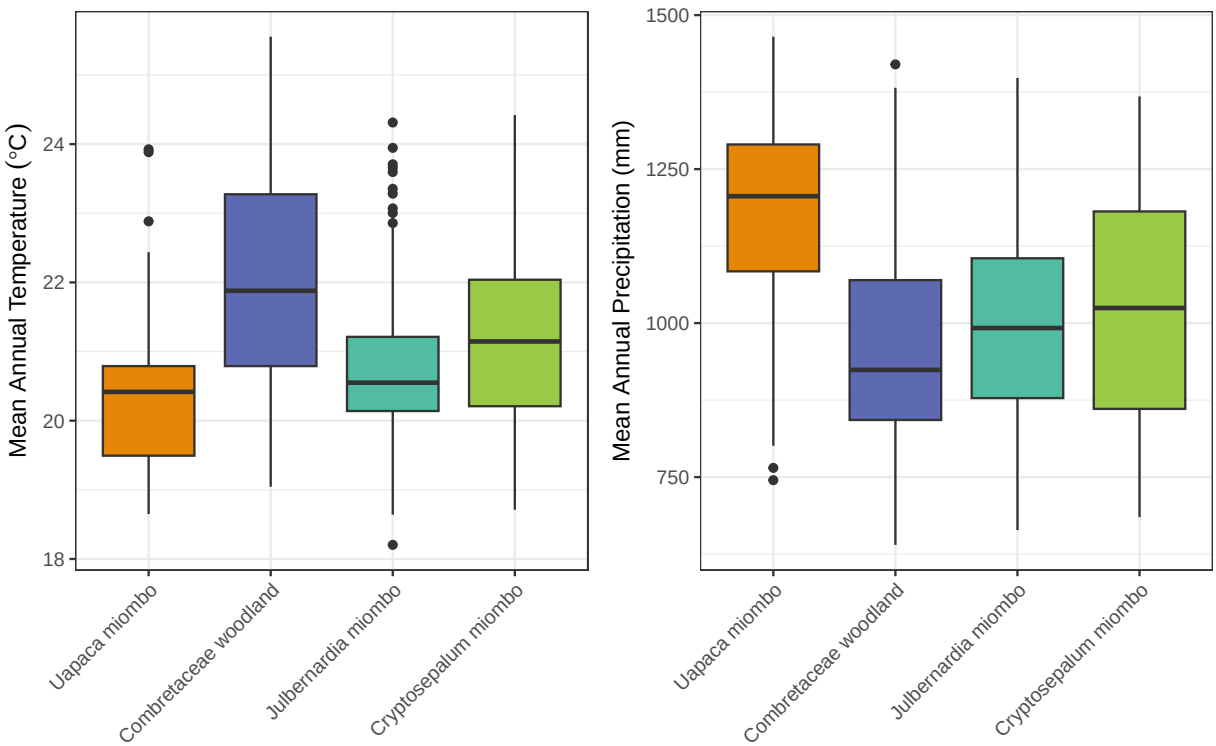

Figure 3: Mean Annual Precipitation and Mean Annual Temperature for each site, grouped by vegetation type. Climate data were extracted from the WorldClim dataset at 30 arc second resolution (Fick & Hijmans, 2017).

To describe species diversity in each site, we calculated the Shannon-Wiener index (H') from species basal area rather than individual abundance, as a measure of species diversity effectively weighted by a species' contribution to canopy occupancy and thus the land surface phenological signal. H' was transformed to the first-order numbers-equivalent (^1D) of H', calculated as $e^{H'}$, also known as effective species richness (Jost, 2007). We use ^1D as the primary measure of species diversity in our statistical models, and is subsequently referred to as species diversity. To describe average tree size, we calculated the quadratic mean of stem diameters per site (Curtis & Marshall, 2000). The quadratic mean of diameter is directly related to basal area, unlike the arithmetic mean of diameter. It is thus a more appropriate descriptor of average tree size within a site, as it scales linearly with total volume and biomass.

2.3 Remote sensing data

Precipitation time series data for each site was compiled from the GPM IMERG Final Precipitation L3 1 day V06 product, which has a pixel size of 0.1°(11.1 km at the equator) (Huffman et al., 2015). We used all available data from 2001 to 2021. There is no clear consensus on best practice for defining the limits of the rainy season (Guan et al., 2014). Here we followed the example of

Stern et al. (1981) and Adole et al. (2018b), variations of which have been used in several studies of African savanna-woodland phenology (Ryan et al., 2017; Tadross et al., 2005; Mupangwa et al., 2011; Segele & Lamb, 2005). We first defined "hydrological years" for each site, bounded by the calendar months experiencing the lowest precipitation in each calendar year (Ferijal et al., 2022). Where multiple months tied for lowest precipitation, the month was chosen which would produce the shortest hydrological year. Note that hydrological years may be longer or shorter than a calendar year.

The onset of the rainy season for each hydrological year was defined as the start of the first period of 10 days with more than 20 mm total rainfall, immediately followed by 20 days with more than 20 mm total rainfall. With this rolling window approach we sought to identify the first sustained period where tree available soil moisture was increasing, i.e. where the average regional evaporation rate of ~1 mm per day was exceeded by rainfall (Campbell, 1996), while accounting for the stochastic nature of rainfall in our study at the start of the rainy season. We chose to relax the rainfall constraint for the subsequent 20 day period to account for variation in the temporal consistency of rainfall at the start of the rainy season in our study region. As opposed to a percentile of cumulative rainfall approach, our method is not affected by dry season storm-burst rainfall events, which, while contributing to overall cumulative rainfall, often do not have the capacity to activate tree foliage production due to their transience (February & Higgins, 2016).

The end of the rainy season was defined as the start of the first period where fewer than four days within a 30 day period received more than 0.5 mm rainfall. Guan et al. (2014) compares other methods of estimating rainy season onset and end dates, and finds little functional difference between the method chosen here and other common methods. To validate our chosen definition of rainy season we checked that all season start and end dates occurred within 10 days of the dates where 10% and 95% of yearly cumulative rainfall were exceeded, for each hydrological year (Adole et al., 2018b).

To quantify seasonal patterns of leaf production at each site, we used the MODIS MCD12Q2 v6.1 land surface phenology product Friedl et al. (2022). MCD12Q2 provides annual metrics describing land surface phenology, with a spatial resolution of ~500 m. The phenological metrics in MCD12Q2 are derived from the 2-band MODIS EVI time series, with nadir BRDF- (Bidirectional Reflectance Distribution Function) adjusted surface reflectance. There are a number of land surface phenology products derived from MODIS vegetation indices. MCD12Q2 was chosen here as it has been used effectively in other studies in deciduous African woodlands to define the growing season (Begue et al., 2014; Adole et al., 2018a). Additionally, a previous comparison of MODIS-derived land surface phenology products showed that MCD12Q2 out-performed other products in predicting the start of the growing season in relation to ground observations (Peng et al., 2017).

All sites were determined to have a single annual growing season according to the MCD12Q2 "Num-Cycles" metric. We used all years of data from 2001 to 2021. We used the "Greenup" and "Dormancy" metrics to define the growing season as the period when the EVI time series exceeds 15% of the EVI amplitude for a given growing season (Figure 4). We used the "Maturity" metric to define the end of the green-up period, and the "Senescence" metric to define the start of the senescence period. We used these dates to calculate the length of the green-up period and senescence period. We calculated "pre-rain green-up" as the number of days between the onset of the growing season and the onset of the rainy season, for each year of available data. Similarly we calculated "senescence lag" as the number of days between the end of the rainy season and the end of the growing season. To aid interpretation of statistical models, we reversed the sign of the pre-rain green-up measurements, so that larger values indicate earlier pre-rain green-up. As senescence generally occurred after the end of the rainy season, we did not reverse the sign of senescence lag, so larger values indicate later senescence lag.

Daily total precipitation from the GPM IMERG product was separated into three periods: precipitation during the growing season, precipitation in the 30 day period before the onset of the growing season (pre-green-up precipitation), and precipitation in the 30 day period before the onset of senescence at the end of the growing season (pre-senescence precipitation).

Temperature time series data for each site was compiled from the ERA5-Land hourly data, using the 2 m air temperature variable at 0.1°(11.1 km at the equator) (Muñoz-Sabater et al., 2021). We aggregated this data to the maximum temperature for each day over the period 2001 to 2021. We then calculated cumulative temperature during the 30 day pre-green-up period, 30 day pre-senescence period, and the growing season period, as with the precipitation data. Cumulative temperature has been shown in previous studies to be an important cue driving patterns of leaf production in more temperate systems (Archibald & Scholes, 2007; Michelson et al., 2017; Gárate-Escamilla et al., 2020).

The spatial scales of the precipitation data (0.1°) and land surface phenology data (500 m) do not match. We do not see this as a problem however, as we do not rely on individual localised rainfall events to derive precipitation metrics. Instead we calculate averaged values over a given period, e.g. 30 days prior to green-up.

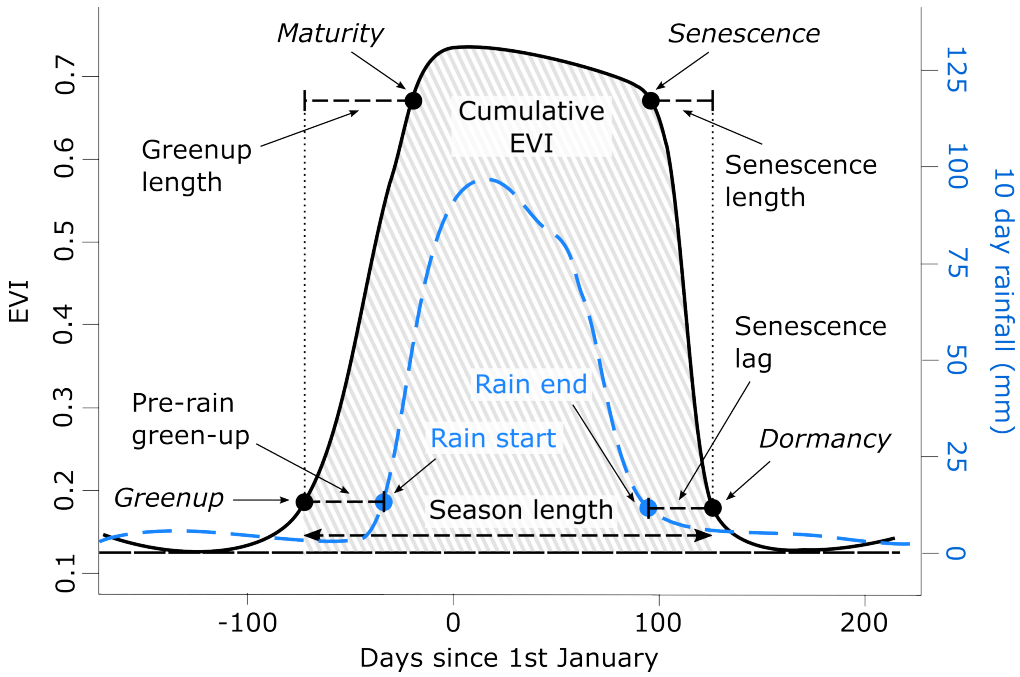

Figure 4: Schematic diagram of hypothetical splined NBAR-EVI2 time series (black curve), from which metrics provided by the MCD12Q2 product are derived (black circles). Cumulative EVI indicates the shaded area under the EVI curve for a given growing season, bounded by the "Greenup" and "Dormancy" dates. The blue curve shows a hypothetical splined 10 day rolling total rainfall time series, derived from the GPM IMERG product. Blue circles indicate the start and end of the rainy season as defined by our algorithm. Dashed black lines indicate the derived phenological metrics used in this study. In this hypothetical example, "Greenup" occurs in advance of the start of the rainy season, and "Dormancy" occurs after the end of the rainy season, reflecting the condition of 72.7% (8451/11625) of the yearly data in our dataset. In 26.7% (3106/11625) of cases, senescence occurred after the end of the rainy season, but green-up occurred after the start of the rainy season. Adapted from (Gray et al., 2022).

2.3.1 Statistical modelling

We used multivariate linear mixed effects models to investigate the effect of tree species diversity, species composition, and woodland structure on each of the six phenological metrics. We defined a maximal model structure including the explanatory variables of species diversity, mean tree stem diameter, and relative abundance of Detarioideae species, alongside climatic variables shown by previous studies to strongly influence patterns of foliage production and senescence: precipitation and cumulative temperature. For models of growing season length and cumulative EVI we used total cumulative rainy season precipitation and total cumulative rainy season temperature; for pre-rain green-up and length of green-up period we used pre-green-up precipitation and cumulative temperature; for senescence lag and length of senescence period we used pre-senescence precipitation

and cumulative temperature. We included interaction terms for species diversity and mean stem diameter with vegetation type, to test whether variation in community composition among vegetation types mediated the effect of these variables on land surface phenology. All models included a random intercept term for site to account for yearly repeated measurements of site land surface phenology. We used the `dredge` function from the `MuMIn` R package to determine which combination of explanatory variables and their interactions best explained each phenological metric. `dredge` produces a suite of models containing all possible combinations of explanatory variables and their defined interactions. Candidate models were compared using the model log-likelihood, AIC (Akaike Information Criteria), conditional R^2. Where similar models were within two AIC points of each other, the model with fewer terms was chosen as the best model, to maximise model parsimony. Explanatory variables in each model were standardised to Z-scores prior to modelling to allow comparison of slope coefficients within a given model.

We assessed standardised effect sizes for explanatory variables in each best model, re-estimated using Maximum Likelihood, to determine the magnitude and direction of their effect on each phenological metric. For each explanatory variable we calculated the 95% confidence interval of the effect size as 1.96× the standard error of the parameter estimate (Zuur et al., 2010). To investigate variation in the role of species diversity and woodland structure on phenological metrics among vegetation types, we estimated marginal means for these explanatory variables for each of the vegetation types, if they appeared in the best model. We estimated marginal means using the `ggeffects` package in R. Estimating marginal means entails generating model predictions across values of a focal variable, while holding non-focal variables constant at their reference value. We estimated marginal means across the full range of each predictor in our dataset (species richness: 1-20, mean stem diameter: 10-35 cm).

To identify significant variation in land surface phenology among vegetation types we conducted a MANOVA using site median phenological metrics across the study period as response variables, followed by post-hoc Tukey's tests between each pairwise combination of vegetation types per phenological metric, to test whether vegetation types differed significantly in their land surface phenology. All statistical analyses were conducted in R version 4.0.2 (R Core Team, 2020).

3 Results

Multivariate linear mixed effects models effectively explained pre-rain green-up (R^2_m=0.46, R^2_c=0.65), cumulative EVI (R^2_m=0.22, R^2_c=0.72), season length (R^2_m=0.42, R^2_c=0.63) and green-up length (R^2_m=0.2, R^2_c=0.38), while senescence length (R^2_m=0.01, R^2_c=0.2) and senescence lag (R^2_m=0.03, R^2_c=0.29), were poorly constrained even in the best models, according to the model selection proce-

dure. Fixed effects explained a large proportion of the total variance explained in models for pre-rain green-up, season length and green-up length, while the random effect of site explained most of the variance in cumulative EVI.

We compared standardised slope coefficients of explanatory variables within the best model for each phenological metric to determine the strength and direction of their influence (Figure 5, Table S1). Tree species diversity exhibited strong positive effects on season length (H_1) and pre-rain green-up (H_2). In contrast, species diversity exhibited a strong negative effect on senescence lag, though this model explained little of the total variance in senescence lag. Average tree size, measured by the quadratic mean of stem DBH per site, was a significant predictor of pre-rain green-up and green-up length (H_3). The relative basal area abundance of Detarioideae species was a significant predictor of pre-rain green-up, season length and senescence lag (H_4).

The slope of relationships between diversity and structural metrics and phenological metrics varied little among vegetation types (Figure 6). The strong positive effect of species diversity on season length and pre-rain green-up was shared among all vegetation types. The effect of mean stem diameter on pre-rain green-up and green-up length varied markedly by vegetation type, with a negligible effect of this variable on green-up patterns in Cryptosepalum miombo and Julbernardia miombo, but stronger positive effects in Uapaca miombo and Combretaceae woodland.

A MANOVA using site median values for each phenological metric, showed a significant difference among vegetation types ($F(3,615)=10.76$, $p<0.01$). Combretaceae woodlands had a significantly shorter growing season length than the other vegetation types, along with significantly shorter periods of green-up (Figure 7). The majority of plots, regardless of vegetation type, exhibited some degree of pre-rain green-up, with some inter-annual variation. Across all sites, the mean lag between green-up and the onset of seasonal rains was 14±20.2 days (±1 standard deviation).

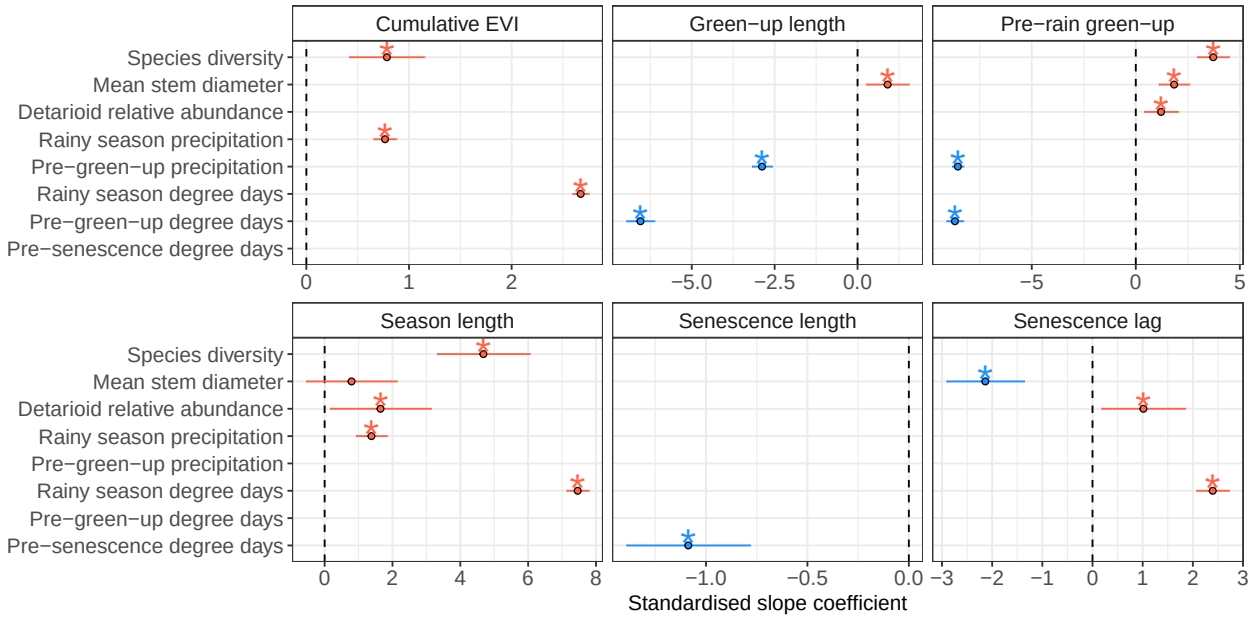

Figure 5: Standardised slope coefficients for the best model of each phenological metric. Slope estimates are ±95% confidence interval. Slope estimates where the interval does not overlap zero are considered to be significant effects and are marked by asterisks.

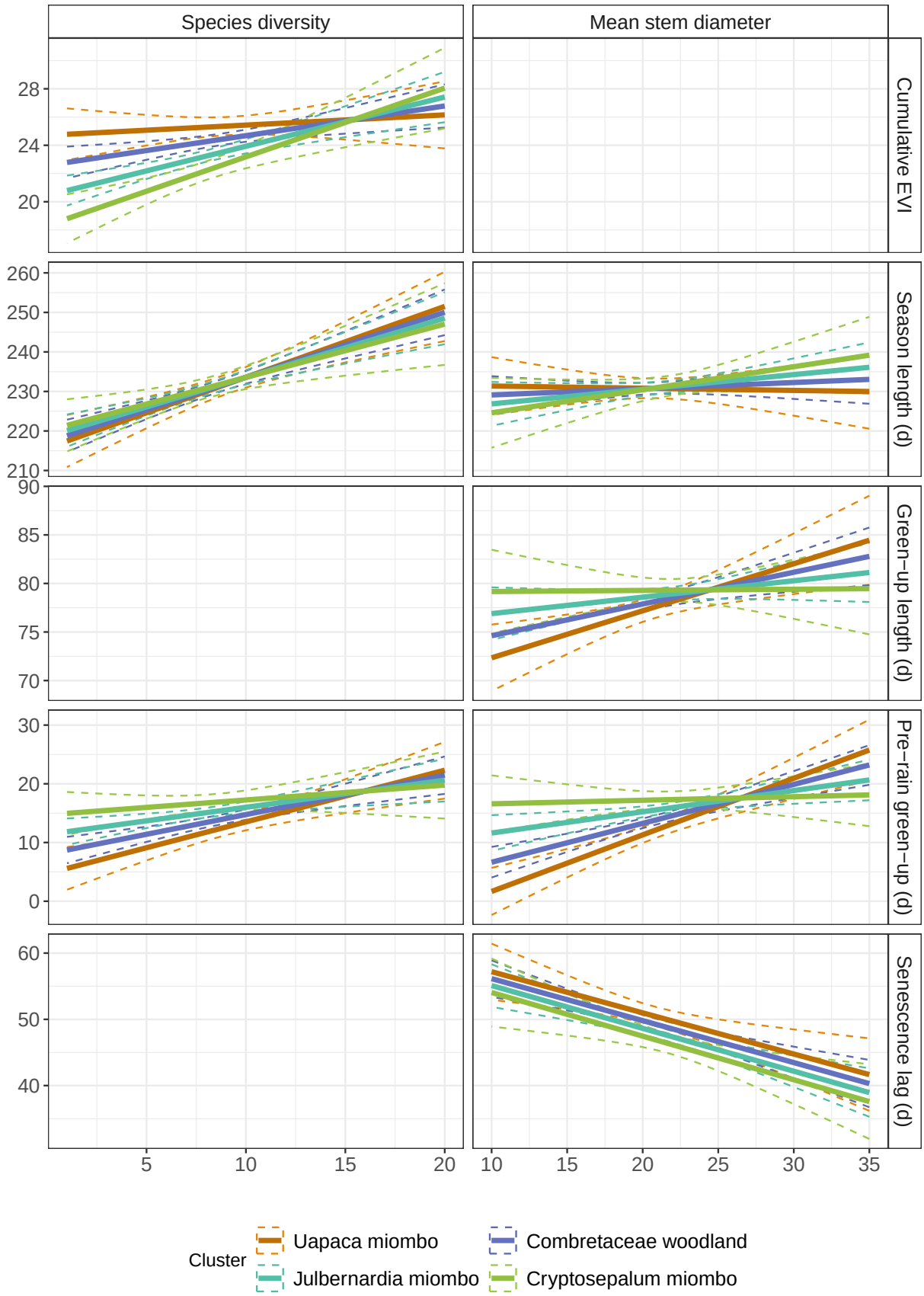

Figure 6: Marginal effects of tree species diversity on each of the phenological metrics, from the best-fitting multivariate linear model, for each vegetation type. Dotted lines represent 95% confidence intervals for each vegetation type. Units of 'd' are number of days.

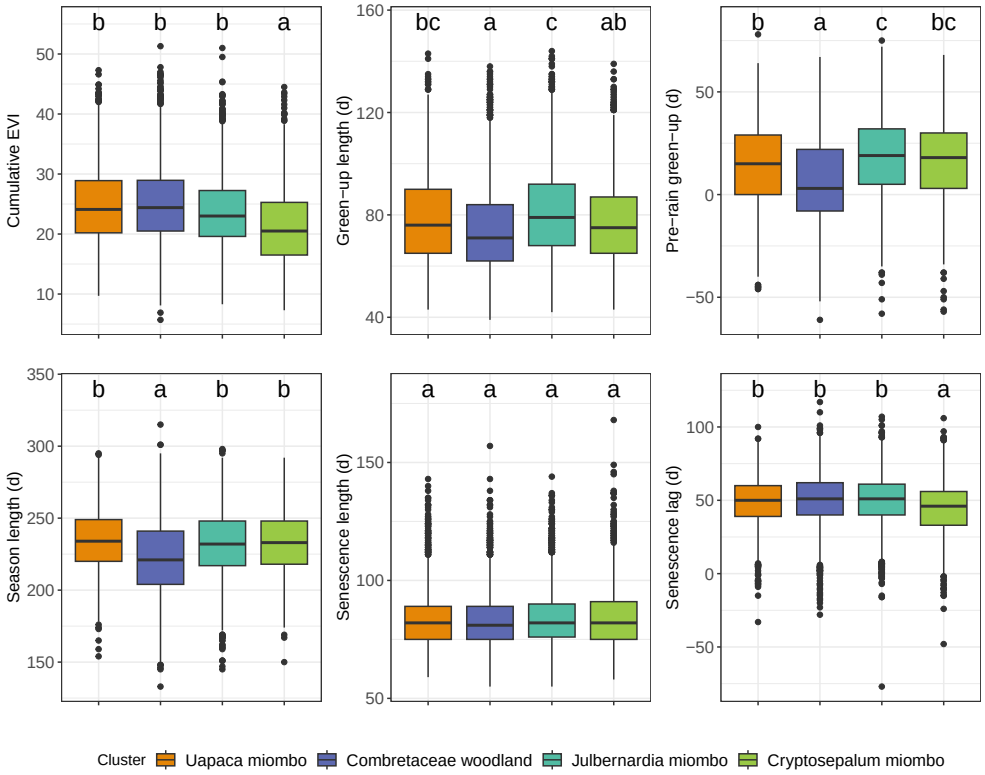

Figure 7: Boxplots of the six phenological metrics used in the study, grouped by vegetation type. Letters not shared among vegetation types denote that these vegetations types were identified as significantly different by post-hoc Tukey's HSD tests derived from the MANOVA of phenological metrics by vegetation type. Units of 'd' are number of days.

4 Discussion

We have demonstrated clear and measurable effects of tree species diversity, composition and structure on various metrics of land surface phenology in deciduous woodlands across Zambia. Tree species diversity was associated with a longer growing season and with earlier pre-rain green-up across woodland compositional types. We also find that sites with a greater basal area abundance of trees from the Detarioideae (subfamily of Fabaceae) tend to green-up earlier in advance of the rainy season. Together, these results imply that at least some of this species diversity effect may be driven by the increased likelihood of containing species able to green-up earlier, i.e. a mass-ratio effect (**Grime1996**). Our study lends support for a general positive effect of tree species diversity on primary productivity in southern African deciduous woodlands.

Our finding that species diversity was associated with earlier pre-rain green-up suggests that diverse woodlands may be more resilient to temporal variations in the onset of seasonal rainfall. The phenomenon of pre-rain green-up provides early forage for herbivores (Morellato et al., 2016), and

provides facilitative effects such as cover and hydraulic lift which benefit understorey plants (Domec et al., 2010; Yu & D'Odorico, 2015). Our results highlight the role of tree species diversity as a driver of key ecosystem processes, affecting ecosystem structure, the wildlife provisioning role, and Gross Primary Productivity (GPP) (Richardson et al., 2009).

The basal area abundance of Detarioideae species had a positive effect on pre-rain green-up. Given their tendency to create deep root networks (Zhou et al., 2020), and their conservative leaf economic spectrum traits, which increase the resilience of leaves to environmental stresses such as cold and drought (Wigley et al., 2016), we suggest that compared to other co-occurring taxonomic groups, Detarioideae species are better equipped to green-up before seasonal rainfall begins. While at present this ability to green-up prior to the rainy season provides a competitive advantage to Detarioideae species, it is unclear whether this behaviour might also lead to increased mortality through hydraulic failure under a changing climate, where the date of onset of the rainy season is predicted to become more erratic (Vinya et al., 2018). Ultimately, it is not well defined how close these dominant species are to the edge of their climatic niche (Jinga & Ashley, 2019). Given the dominance of Detarioideae species in deciduous woodlands across southern Africa, increased mortality of this group may affect the future structure and composition of these woodlands, with consequences for patterns of land surface phenology and therefore woody productivity.

Environmental variables had consistently strong effects on phenological metrics, and confirm previous research. Greater rainfall within the rainy season led to a longer growing season and greater cumulative EVI, matching our expectations that increased water availability allows for greater foliage production and builds groundwater reserves allowing for a longer growing season. Greater rainfall in the 30 days leading up to the start of the growing season was associated with later green-up with respect to the timing of the onset of the rainy season. We suggest this may be the result of large amounts of rainfall kick-starting foliage production in the absence of a pre-rain green-up effect in some plots, and may represent cases where grasses drive the start of growing season land surface phenology signal. Hotter days led to shorter periods of senescence and green-up. In the case of green-up length, this effect may imply an increasing temperature trigger on foliage production, as is seen in temperate woodland ecosystems (Flynn & Wolkovich, 2018). In the case of senescence, this effect may be due to hot days causing wilting of old leaves under reduced water availability at the end of the rainy season (Warren et al., 2011; Mariën et al., 2019). This study provides a glimpse into how extreme climatic events, particularly heat and drought events, could strongly affect leaf phenology in the future under a changing climate in southern Africa (Naumann et al., 2018). Increased temperature and decreased precipitation were both strongly associated with decreased growing season length, and reduced pre-rain green-up. Future studies would be well-placed to explore further how climate extremes affect land surface phenology, ideally in combination with measurements of

composition and structure, which we expect might mediate the effect of climate extremes.

The lag between the end of the growing season and the end of the rainy season was negatively influenced by tree mean stem diameter, though our model had low explanatory power. Cho et al. (2017) similarly found that tree cover, measured by MODIS Leaf Area Index (LAI) data, had a significant negative effect on senescence rates in woodlands in South Africa, which have similar climatic conditions to the sites in our study. In most woodlands, including sparse savannas, while the onset of the growing season is often driven by tree photosynthetic activity (Ryan et al., 2017; Archibald & Scholes, 2007; Chidumayo, 2001), which may precede the onset of precipitation, the end of the growing season is conversely driven by the understorey grass layer (Cho et al., 2017; Guan et al., 2014). Grass activity is more reactive to short-term changes in soil moisture than tree activity, and may oscillate within the senescence period (Archibald & Scholes, 2007). This may explain the lack of a strong precipitation or diversity signal for senescence lag and length of the senescence period in our models.

Our best models predicting the length of the green-up and senescence period did not explain much of the variance in these phenological metrics. Length of the green-up period varied significantly among vegetation types, implying that the particular combination of species and the synchrony of their phenological strategy might be the determining factor for this phenological metric. Though more work must be done to explore this further. As mentioned above, short term oscillations in soil moisture and the condition of ground cover vegetation may be responsible for driving the length of the senescence period, hence the wide variability of this phenological metric and the poor fit of our statistical models. Indeed, most of the variance explained by our model of senescence length and senescence lag came from the random effect of site, implying some unmeasured inter-site variation, possibly in community composition.

It is striking that the fixed effects in our models were unable to explain variation in the length of the senescence period or the lag between the end of the growing season and the end of the rainy season. The random effect of site accounted for most of the variance explained by these models, implying high variation among sites as compared to within sites. This suggests that there are some unmeasured environmental or biotic factors at play in determining senescence in these systems.

Other studies, both globally and within southern African woodlands, have largely ignored patterns of senescence, instead focussing on patterns of green-up (Gallinat et al., 2015). Most commonly, these studies simply correlate the decline of rainfall with senescence (Stevens et al., 2016; Guan et al., 2014), but our best model suggests that temperature is a stronger determinant of the end of the growing season than precipitation. We suggest that hotter days in the period leading up to the start of the senescence period might accelerate senescence due to wilting, resulting in a shorter senescence period length. In temperate ecosystems, which experience autumn senescence, lower night

time temperatures have been shown to increase the rate of senescence and thus decrease senescence period length (Michelson et al., 2017; Gárate-Escamilla et al., 2020). In these water-limited and hot systems however, it seems hotter temperatures are the main driver. Similarly, hotter days prior to senescence appear to advance the onset of senescence with respect to the end of the rainy season, which may also be caused by wilting as rainfall starts to decrease. These suggestions however, must be considered in the context that the fixed effects in our models did not account for much of the variation in senescence period length or senescence lag.

Alternatively, Zani et al. (2020) suggests that in resource-limited environments, senescence times may largely be set by the preceding photosynthetic activity and sink-limitations on growth. For example, limited nutrient supply may prohibit photosynthesis late in the season if the preceding photosynthetic activity has depleted that supply. Reich et al. (1992) suggested that there are many direct constraints on leaf life-span such as drought and herbivory, especially in the seasonally dry tropics, which would lead to timing of senescence being set largely by the time of bud-burst. Our study indirectly corroborates this theory, with the finding of a moderate positive correlation between pre-rain green-up and senescence lag across all vegetation types, indicating that later green-up allows later senescence (Figure S1).

We found positive effects of mean tree size (site quadratic mean diameter) on pre-rain green-up and green-up length, but negative effects on senescence lag. We suggest that the increased rooting depth of large trees, allows them to access deep ground-water reserves, conferring resilience to inter-annual variation in patterns of seasonal rainfall (Holdo et al., 2017). This result provides a possible mechanism by which certain groups of species, like those of the subfamily Detarioideae, are able to drive earlier pre-rain green-up, as they generally grow to larger sizes. Our results suggest a mass-ratio effect in the pre-rain green-up of Zambian deciduous woodlands (Grime, 1998), whereby dominant functional groups of trees which grow to large sizes are able to green-up in advance of seasonal rains, drive longer growing seasons, and overall drive greater productivity.

Our finding that species diversity increases season length and cumulative EVI in seasonally dry tropical woodlands provides an example of a mechanism by which species diversity increases ecosystem function, a pattern observed previously in other studies of southern African woodlands (Godlee et al., 2021; McNicol et al., 2018; Shirima et al., 2015), and globally in other biomes (Plas, 2019; Tilman et al., 2014). Our findings provide earth surface system modellers with a means to better understand how changes in the inter-related variables of species diversity, composition and structure affect land surface phenology. Incorporating predictions of biotic composition and its change over time into earth system models has been limited until recently for two main reasons (Ahlstrom et al., 2015; Bodegom et al., 2011). Firstly, there are large uncertainties in the effects of biodiversity on ecosystem function, e.g. gross primary productivity (GPP). Secondly, until recently there

has been a major paucity of ground data on community composition. Our study comes at a pertinent time when plot-based vegetation monitoring networks have become sufficiently widespread (ForestPlots.net et al., 2021; SEOSAW, 2020), and remotely sensed proxies of diversity in forested ecosystems are becoming sophisticated enough (Schneider et al., 2017; Cavender-Bares et al., 2020) to be of real use to earth system modellers. Our study provides a link by demonstrating a strong relationship between species diversity and land surface phenology, particularly growing season length, which is itself closely correlated with GPP (Sjöström et al., 2011).

5 Conclusion

We explored how tree species diversity, composition and structure of deciduous woodlands influence land surface phenology across Zambia. We showed that species diversity drives longer growing seasons and promotes earlier pre-rain green-up across all woodland compositional types studied here. The abundance of Detarioideae species (subfamily of Fabaceae) had a consistent positive effect on pre-rain green-up and over-all growing season length, supporting suggestions from previous studies of their keystone role in driving pre-rain green-up in southern African miombo woodlands. We found an effect of mean tree size on land surface phenology, and it is likely that larger potential tree size in Detarioideae species is one trait facilitating pre-rain green-up. Finally, we have demonstrated variation in phenological patterns among woodland compositional types within Zambia that are not commonly distinguished in classifications of woodland types in the region.

Our results lend further support to an already well established corpus of study on the positive effect of species diversity on ecosystem function. Additionally, our results provide evidence for earth-system modellers, demonstrating the importance of community composition, structure and diversity in determining patterns of land surface phenology. As reliable data on plant community composition and structure becomes available across larger spatial scales and at higher spatial resolution, we hope that these data can be included in models of land-atmosphere mass and energy exchange to improve their predictive power.

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6 Supplementary Material

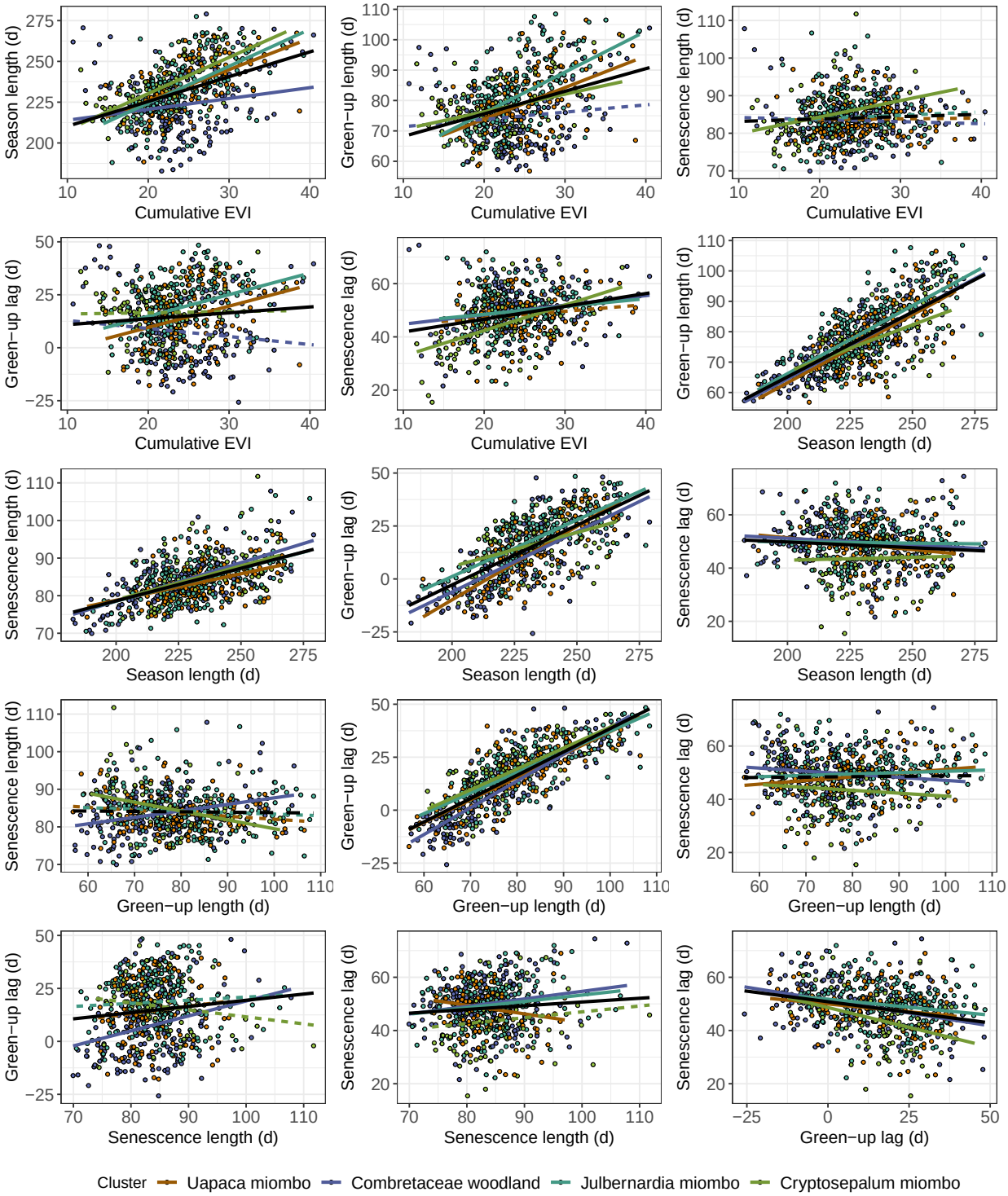

Figure S1: Scatter plots showing pairwise comparisons of the six phenological metrics used in this study, extracted from the MODIS MCD12Q2 product (Friedl et al., 2022). Points represent study sites and are coloured by vegetation type. Linear regression line of best fit for all sites is shown as a black line, while linear regressions are shown for each vegetation type as coloured lines. Non-significant linear regression fits ($p<0.05$) have dashed lines. The units of 'd' are the number of days.

Table S1: Performance of the top five candidate models for each phenological metric, showing parameter estimates (

pm95% confidence interval), and goodness of fit statistics. Models are ranked by AIC values. $R^2{}_m$ and $R^2{}_c$ refer to the marginal and conditional R^2, respectively. The overall best model is marked by bold text, according to AIC and model parsimony. Parameter estimates are marked by an asterisk where thei 95% confidence interval does not overlap zero.

Response	Rank	Precipitation	Temperature	Detarioid BA	Stem diameter	Richness	DoF	AIC	R^2_m	R^2_c	W_i
Cumulative EVI	**1**	**0.77(0.111)***	**2.67(0.078)***			**0.78(0.365)***	**7**	**58129**	**0.22**	**0.72**	**0.585**
Cumulative EVI	2	0.77(0.111)*	2.67(0.078)*	-0.26(0.412)		0.72(0.38)*	8	58131	0.22	0.72	0.242
Cumulative EVI	3	0.77(0.111)*	2.67(0.078)*		0.1(0.369)	0.8(0.369)*	8	58132	0.22	0.72	0.116
Cumulative EVI	4	0.77(0.111)*	2.67(0.078)*	-0.28(0.416)	0.13(0.373)	0.73(0.382)*	9	58134	0.22	0.72	0.054
Cumulative EVI	5	0.78(0.111)*	2.67(0.078)*	-0.48(0.399)*			7	58141	0.21	0.72	0.002
Season length	**1**	**1.38(0.456)***	**7.46(0.325)***	**1.65(1.488)***	**0.79(1.332)**	**4.68(1.368)***	**9**	**88227**	**0.14**	**0.63**	**0.496**
Season length	2	1.39(0.456)*	7.47(0.325)*	1.77(1.473)*		4.6(1.361)*	8	88227	0.14	0.63	0.400
Season length	3	1.38(0.456)*	7.46(0.325)*		1(1.323)	4.28(1.324)*	8	88231	0.14	0.63	0.068
Season length	4	1.39(0.456)*	7.46(0.325)*			4.14(1.311)*	7	88232	0.14	0.63	0.036
Season length	5		7.28(0.319)*	1.72(1.548)*	0.87(1.386)	5.12(1.416)*	8	88257	0.13	0.64	0.000

Continued overleaf

Table S1: (continued)

Response	Rank	Precipitation	Temperature	Detarioid BA	Stem diameter	Richness	DoF	AIC	R^2_m	R^2_c	W_i
Green-up length	**1**	**-2.88(0.301)***	**-6.55(0.42)***		**0.91(0.639)***		**7**	**86430**	**0.20**	**0.38**	**0.320**
Green-up length	2	-2.89(0.302)*	-6.59(0.425)*		0.84(0.645)*	-0.44(0.65)	8	86431	0.20	0.38	0.237
Green-up length	3	-2.9(0.302)*	-6.61(0.425)*	-0.5(0.727)	0.9(0.651)*	-0.57(0.674)	9	86431	0.20	0.38	0.202
Green-up length	4	-2.88(0.301)*	-6.55(0.421)*	-0.34(0.701)	0.96(0.649)*		8	86432	0.20	0.38	0.164
Green-up length	5	-2.91(0.301)*	-6.59(0.425)*			-0.56(0.645)	7	86435	0.20	0.38	0.030
Senescence length	**1**		**-1.09(0.305)***				**5**	**81653**	**0.01**	**0.20**	**0.206**
Senescence length	2	0.23(0.234)*	-0.99(0.32)*				6	81654	0.01	0.20	0.157
Senescence length	3		-1.08(0.305)*		-0.32(0.469)		6	81654	0.01	0.20	0.112
Senescence length	4	0.24(0.234)*	-0.98(0.32)*		-0.32(0.468)		7	81655	0.01	0.20	0.086
Senescence length	5		-1.06(0.309)*			0.24(0.473)	6	81655	0.01	0.20	0.076
Pre-rain green-up	**1**	**-8.55(0.268)***	**-8.69(0.398)***	**1.2(0.821)***	**1.84(0.735)***	**3.72(0.758)***	**9**	**83903**	**0.46**	**0.65**	**0.559**
Pre-rain green-up	2	-8.55(0.268)*	-8.68(0.398)*	1.3(0.82)*	1.81(0.727)*	3.44(0.73)*	8	83905	0.46	0.65	0.281
Pre-rain green-up	3	-8.53(0.268)*	-8.63(0.395)*		1.93(0.722)*		7	83907	0.46	0.65	0.117
Pre-rain green-up	4	-8.53(0.268)*	-8.63(0.395)*	1.2(0.794)*	1.93(0.733)*		8	83909	0.46	0.65	0.044
Pre-rain green-up	5	-8.56(0.268)*	-8.7(0.399)*			4.48(0.735)*	7	83925	0.45	0.65	0.000

Continued overleaf

Table S1: (continued)

Response	Rank	Precipitation	Temperature	Detarioid BA	Stem diameter	Richness	DoF	AIC	R^2_m	R^2_c	W_i
Senescence lag	**1**		**2.4(0.324)***	**1.01(0.833)***	**-2.14(0.771)***		**7**	**88232**	**0.03**	**0.29**	**0.420**
Senescence lag	2		2.39(0.324)*	0.93(0.864)*	-2.17(0.775)*	-0.3(0.792)	8	88233	0.03	0.30	0.205
Senescence lag	3	0.21(0.434)	2.43(0.329)*	1.02(0.836)*	-2.14(0.773)*		8	88234	0.03	0.30	0.136
Senescence lag	4	0.25(0.44)	2.43(0.329)*	0.92(0.868)*	-2.18(0.778)*	-0.38(0.807)	9	88235	0.03	0.30	0.079
Senescence lag	5		2.39(0.324)*		-1.98(0.762)*		6	88236	0.03	0.29	0.063